HackerFab IITB Recruitment Tasks

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Github repo: https://github.com/suresh-vlsi/KSureshKumar_HackerFabIITB_Submission

Design of a Hot Plate for Semiconductor Processing

Problem Statement

Develop an end-to-end design including a BOM and assembly instructions for a Hot Plate that meets the following specifications:

1. It can process a 4" wafer - the part of the plate which meets specs 2-5 should contain a minimum diameter of 4".
2. The plate should allow setting to a temperature range from the room temperature in the cleanroom to a maximum of 400°C.
3. The set temperature to actual temperature error should be <5°C. What's more important in materials processing is run-to-run repeatability. The temperature from one heating run to the next one should differ by less than 0.5°C. How would you ensure this? (Hint: The answer can be a process or a tool solution, usually you'll need both.)
4. The uniformity of temperature across the 4" wafer should be within ±0.5°C. That is, there should be no two points on the wafer whose temperature will vary by more than 1°C.
5. The top surface will need to be machined flat to a tolerance of ±10 μm.

End-to-End Design of a Hot Plate for Semiconductor Processing

1. Design Overview

The objective of this project is to design a temperature-controlled hot plate suitable for semiconductor processing using 4-inch silicon wafers. The proposed system must operate from cleanroom ambient temperature to 400°C while maintaining temperature accuracy, run-to-run repeatability, wafer-area temperature uniformity, and surface flatness.

The proposed design consists of a 160 mm diameter aluminium heat-spreading plate, four 250 W cartridge heaters, two independently controlled heating zones, Pt100 resistance temperature detectors (RTDs), a closed-loop temperature controller, thermal insulation, and an independent over-temperature protection system.

The design process includes mechanical design, thermal calculations, heater selection, temperature sensing, feedback control, electrical safety, component selection, assembly, commissioning, and experimental validation.

The proposed design is a prototype concept. Compliance with the specified temperature and flatness tolerances must be demonstrated through physical measurements.

2. Mechanical Design

2.1 Heating Plate

The proposed heating plate has the following nominal dimensions:

- Material: Aluminium 6061, subject to confirmation of suitability for prolonged exposure at 400°C.
- Finished diameter: 160 mm.
- Finished thickness: 20 mm.
- Intended wafer diameter: 101.6 mm (4 inches).
- Surface finish: Final-ground top surface.
- Target flatness: ±10 μm, subject to measurement and an agreed tolerance definition.

The 160 mm plate diameter provides sufficient geometric area for a 4-inch wafer. The aluminium plate acts as a heat spreader, distributing thermal energy from the cartridge heaters towards the wafer-supporting surface.

However, aluminium alloy properties change at elevated temperatures. The plate may lose its original heat-treated temper or distort during repeated heating cycles. Therefore, the material condition, machining sequence, thermal conditioning, and final surface grinding must be reviewed with a qualified workshop.

The finished plate must be inspected after machining and again after thermal cycling. The flatness requirement must not be considered satisfied until the measurement results demonstrate compliance.

2.2 Heater Arrangement

Four cartridge heaters, each rated at 250 W and 230 V, are proposed. The total nominal installed heating power is:

Total heater power = 4 × 250 W = 1,000 W.

The heaters are divided into two independently controlled zones:

- Centre zone: Two cartridge heaters, providing 500 W nominal power.
- Edge zone: Two cartridge heaters, providing 500 W nominal power.

The centre zone supplies heat to the central region, while the edge zone allows compensation for heat losses near the plate perimeter. The final heater-hole locations, spacing, depth, insertion length, and retention method must be specified in a dimensioned engineering drawing and verified against the heater manufacturer's requirements.

The two-zone arrangement provides a means of adjusting the temperature distribution. It does not, by itself, guarantee uniformity within ±0.5°C.

2.3 Thermal Insulation and Enclosure

The underside and sides of the heating assembly will be insulated using materials specifically rated for the operating temperature and environment. The insulation reduces unwanted heat losses and helps improve control stability.

The supporting structure and enclosure must:

- Support the plate without excessive thermal distortion.

- Prevent accidental contact with hot surfaces and electrical terminals.
- Keep temperature-sensitive control electronics away from the heated assembly.
- Provide suitable electrical insulation, protective earthing, and strain relief.
- Allow inspection and replacement of heaters and sensors.

The enclosure and insulation arrangement must be reviewed for thermal expansion, heat accumulation, material degradation, and contamination risks.

3. Temperature Measurement and Control System

3.1 Temperature Sensors

Two four-wire Pt100 RTD sensors are proposed. One sensor will monitor the central region, and the second will monitor the edge region within the wafer footprint.

Four-wire measurement reduces the influence of lead-wire resistance. The sensors and measurement electronics must be suitable for the operating temperature and calibrated against a reference instrument.

Two embedded sensors are intended for feedback control. Additional calibrated thermocouples or a thermocouple wafer will be required to map the temperature across the wafer area during validation.

3.2 Closed-Loop Temperature Control

Each heating zone will use an independently adjustable PID temperature-control loop. A microcontroller or equivalent controller will acquire the RTD measurements and adjust heater power through appropriately rated solid-state relays (SSRs).

The control system should provide:

- Adjustable temperature setpoints from cleanroom ambient to 400°C.
- Controlled heating ramps to limit overshoot.
- PID control with appropriate tuning and anti-windup.
- Sensor open-circuit and short-circuit fault detection.
- Temperature and controller-output logging.
- Defined soaking and equilibration procedures.
- Independent hardware over-temperature protection.

The temperature controller must be calibrated and tuned experimentally. Its accuracy cannot be inferred solely from the nominal accuracy of the RTD sensors.

3.3 Independent Safety Protection

Because the system combines mains electricity with temperatures up to 400°C, independent protection is essential.

The proposed safety architecture includes a manual-reset over-temperature cut-out, a suitably rated contactor, correctly selected overcurrent protection, protective earthing, guarded terminals, an emergency stop, and temperature-rated wiring.

The over-temperature protection must be independent of the normal control software. The final trip setting and component ratings must be established through a documented safety review.

Mains wiring, insulation testing, earthing verification, and electrical commissioning must be carried out by a qualified electrical technician.

4. Thermal Design Calculations

The first-order thermal model considers the plate's mass, heat capacity, heating energy, ideal ramp power, and estimated thermal losses.

For a cylindrical plate:

- Volume = $\pi r^2 h$
- Mass = density × volume
- Energy required = mass × specific heat capacity × temperature rise
- Average ideal heating power = energy required / heating time

For the proposed 160 mm diameter and 20 mm thick aluminium plate, the calculated results are:

Parameter	Estimated result
Plate volume	402.1 cm^3
Plate mass	1.086 kg
Plate heat capacity	972.8 J/K
Energy required for heating from 25°C to 400°C	364.8 kJ
Ideal average power for a 20-minute ramp	304 W
Estimated steady-state heat losses at 400°C	172.1 W
Nominal installed heater power	1,000 W

These calculations indicate that 1 kW of installed heater power is a reasonable starting point for the prototype. The actual heating time will depend on heater contact, insulation performance, wafer loading, ambient conditions, and control behaviour.
The calculations do not establish temperature accuracy, wafer-area uniformity, repeatability, or flatness. Those requirements must be verified experimentally.

5. Bill of Materials (BOM)

The following table provides a preliminary bill of materials for the proposed hot plate.

Item	Component	Quantity	Estimated cost (INR)
1	Aluminium 6061 plate, raw stock for Ø160 mm × 20 mm finished plate	1	3,000
2	Machining, heater and sensor holes, thermal conditioning, and surface grinding	1 lot	8,000
3	Cartridge heater, 250 W, 230 V, approximately 6.35 mm diameter	4	2,800
4	Four-wire Pt100 RTD sensors	2	3,000
5	RTD interface modules	2	1,600
6	Microcontroller development board	1	800
7	Appropriately rated zero-cross SSRs and heatsinks	2	1,800
8	Independent manual-reset over-temperature cut-out	1	1,500
9	Contactor, emergency stop, and control-circuit components	1 lot	1,500
10	Low-voltage power supply and regulator	1 lot	1,000
11	High-temperature underside insulation	1 lot	5,000
12	High-temperature side insulation	1 lot	1,000
13	Metal enclosure, standoffs, and earthing hardware	1 lot	4,000
14	Temperature-rated wafer supports	3	1,500
15	High-temperature thermal interface material, if required	1 lot	1,000
16	High-temperature wiring, terminals, fuses, and protective devices	1 lot	2,500
17	Display and rotary encoder	1 lot	1,200
18	Reference temperature instrument and logging allowance	1 lot	4,000
	Total estimated cost		**₹46,700**

All prices are preliminary budget allowances rather than verified supplier quotations. Actual costs depend on availability, specifications, taxes, shipping, calibration requirements, machining tolerances, and rework.
Before purchasing, verify each heater's sheath temperature rating, allowable watt density, electrical resistance, insertion depth, and mounting requirements. Similarly, the temperature rating and accuracy of the sensors, insulation, safety cut-out, and control components must be confirmed from their datasheets.
The budget excludes any additional certification, specialist metrology, or engineering costs not included in the allowances above.

6. Assembly Instructions

I. Step 1: Finalise the Engineering Drawings

Prepare dimensioned drawings for the plate, heater holes, sensor locations, mounting points, insulation, and enclosure. Specify the finished diameter, thickness, flatness definition, and relevant manufacturing tolerances.

Confirm the design with the machining workshop and verify that the proposed heater and sensor dimensions match the selected components.

II. Step 2: Machine the Heating Plate

Obtain the aluminium stock and rough-machine it to the required geometry. Have the workshop determine an appropriate material-conditioning and stress-relief sequence. Complete the final machining and surface grinding only after the agreed conditioning process.

Measure and record the top-surface flatness after machining. The plate should not proceed to final acceptance without a documented measurement.

III. Step 3: Install the Cartridge Heaters

Clean and inspect the heater bores. Install the four cartridge heaters using the manufacturer's approved retention method and specified fit.

Arrange the heaters into the centre and edge zones. Provide strain relief for their electrical leads and ensure that heater connections are protected from heat and accidental contact.

Do not use an improvised thermal compound or retention material unless its temperature rating and electrical suitability have been verified.

IV. Step 4: Install the Temperature Sensors

Install one Pt100 sensor in the central region and the other in the edge region. Ensure that the sensor locations provide representative measurements and that the installation method provides repeatable thermal contact.

Route the sensor wiring separately from mains wiring where practicable, and use suitable shielding and grounding practices to minimise electrical interference.

V. Step 5: Assemble the Insulation and Mechanical Enclosure

Install the approved underside and side insulation. Assemble the supporting structure and enclosure while allowing for thermal expansion and ensuring that insulation does not contact unprotected electrical terminals.

The design must permit inspection, safe maintenance, and replacement of heaters and sensors.

VI. Step 6: Assemble the Control Electronics

Mount the controller, RTD interfaces, SSRs, heatsinks, contactor, over-temperature cut-out, power supply, emergency stop, and protective devices in the designated enclosure.

Keep low-voltage control circuits appropriately separated from mains circuits. Select wiring, terminals, fuses, and protection according to the actual load and applicable electrical requirements.

This stage must be completed and checked by a qualified electrical technician.

VII. Step 7: Perform Electrical Safety Checks

Before energising the system, verify protective-earth continuity, insulation integrity, overcurrent protection, wiring connections, terminal guarding, emergency-stop operation, and independent over-temperature shutdown.

Test sensor fault handling and verify that a detected fault causes heater power to be removed safely. Do not energise an exposed or unreviewed assembly.

VIII. Step 8: Commission the Temperature Controller

Begin commissioning at a modest temperature using a guarded setup. Verify that each sensor responds correctly and that each heating zone responds to its own control command.

Compare the RTD readings against a calibrated reference. Tune each control loop, then test both zones operating together. Use controlled heating ramps and verify that the controller avoids unacceptable overshoot.

IX. Step 9: Perform Thermal Conditioning

Conduct controlled heating and cooling cycles under appropriate supervision. Inspect the heater retention, wiring, insulation, plate condition, and sensor readings.

After thermal cycling, remeasure the plate flatness. Any distortion must be documented and corrected through a revised manufacturing process before claiming compliance.

X. **Step 10: Validate the Finished System**

Perform the temperature accuracy, repeatability, uniformity, and flatness tests described in the following section. Record the raw data, test conditions, calibration information, and pass/fail results.

7. Validation and Acceptance Testing

7.1 Temperature Accuracy

Compare the setpoint with a calibrated reference instrument at representative temperatures, including 100°C, 200°C, 300°C, and 400°C.

The absolute difference between the setpoint and reference temperature must remain below 5°C at every specified test point.

7.2 Run-to-Run Repeatability

Define a standard process recipe, including ramp rate, soak time, wafer placement, support configuration, ambient conditions, and the criterion for thermal equilibrium.

Perform ten independent heating cycles at representative process temperatures. At each measurement location, calculate the mean, standard deviation, and maximum-minus-minimum temperature across runs.

The repeatability requirement is satisfied only when the defined run-to-run temperature difference is below 0.5°C. Measurement uncertainty must be documented.

7.3 Temperature Uniformity Across the Wafer

Use a calibrated thermocouple wafer or a suitably arranged set of calibrated temperature sensors to map the temperature across the full 101.6 mm wafer footprint.

Measure the centre and multiple locations towards the edge after the system reaches a defined steady state.

Calculate:

Temperature span = maximum measured temperature − minimum measured temperature.

The requirement is satisfied only if the temperature span does not exceed 1.0°C across the wafer footprint at the specified test conditions.

Two embedded control sensors alone are insufficient to establish compliance with this requirement.

7.4 Surface Flatness

Measure the finished plate using a suitable calibrated metrology instrument and a documented reference surface. Record measurements at multiple locations across the wafer-supporting area.

Repeat the measurements after controlled thermal cycling. State the exact interpretation of the ±10 μm tolerance and the measurement uncertainty.

If flatness at operating temperature is required, separate high-temperature metrology is necessary; room-temperature measurements alone cannot establish hot-state flatness.

8. Major Engineering Challenges and Mitigation

i. **Plate distortion:** Elevated temperature and thermal cycling may change the plate's geometry. Mitigation includes a qualified material-conditioning process, final grinding after conditioning, and flatness measurements before and after cycling.

ii. **Temperature nonuniformity:** Edge heat losses and uneven heater coupling can create temperature gradients. Mitigation includes independent heating zones, suitable insulation, controller tuning, and thermocouple-wafer mapping.

iii. **Sensor error and drift:** RTD installation, calibration, and electrical interference can introduce errors. Mitigation includes four-wire measurement, reference calibration, controlled sensor placement, and periodic verification.

iv. **Run-to-run variation:** Changes in ramp rate, soak duration, wafer placement, airflow, or sensor condition can change results. Mitigation includes a standardised recipe, logged data, and a defined equilibrium criterion.

v. **Electrical and thermal hazards:** Mains voltage, hot surfaces, and failed switching components create safety risks. Mitigation includes qualified electrical design, independent over-temperature protection, protective earthing, suitable circuit protection, guarded terminals, and controlled commissioning.

9. Conclusion

The proposed hot plate provides a starting architecture for 4-inch semiconductor wafer processing with a nominal operating range from cleanroom ambient to 400°C. The design includes a preliminary mechanical configuration, thermal calculations, a bill of materials, assembly instructions, control-system architecture, safety considerations, and a validation plan.

The first-order calculations support the selection of 1 kW nominal installed heating power as a prototype starting point. However, they do not demonstrate compliance with the strict temperature accuracy, run-to-run repeatability, wafer-area uniformity, or surface-flatness requirements.

The next stage is to finalise the dimensioned drawings, obtain supplier and workshop quotations, complete the safety review, build the prototype, and perform the specified measurements. Only experimentally validated results should be presented as achieved performance.

Solution:

The design uses a 160 mm diameter aluminium plate, four 250 W cartridge heaters, two independently controlled heating zones, and temperature feedback from Pt100 sensors.

Installed heater power **1,000 W** Four 250 W heaters	Target maximum temperature **400°C** Design requirement
Estimated plate mass **1.086 kg** Aluminium, 20 mm thick	Estimated BOM cost **₹46,700** Budget estimate, not a supplier quote

Thermal calculation results:

calcs/thermal_calc.py

```
"""
This is a screening model, not a thermal FEA or measured result.
"""
import math

SIGMA = 5.670374419e-8  # Stefan-Boltzmann constant, W m^-2 K^-4

# Geometry and material assumptions
DIAMETER_M = 0.160
THICKNESS_M = 0.020
RHO_AL6061 = 2700.0      # kg/m^3
CP_AL6061 = 896.0        # J/(kg K)
K_AL6061 = 167.0         # W/(m K), approximate room-temperature value

T_AMBIENT_C = 25.0
T_SET_C = 400.0
EMISSIVITY = 0.20        # assumed; actual surface finish changes this
H_CONVECTION = 8.0       # W/(m^2 K), rough natural-convection assumption
INSULATION_K = 0.03      # W/(m K), assumed insulation conductivity
INSULATION_THICKNESS_M = 0.025
RAMP_MIN = 20.0
WAFER_DIAMETER_M = 0.1016
WAFER_THICKNESS_M = 525e-6
RHO_SILICON = 2330.0
CP_SILICON = 700.0       # approximate J/(kg K)

R = DIAMETER_M / 2
A_TOP = math.pi * R**2
A_SIDE = 2 * math.pi * R * THICKNESS_M
V = A_TOP * THICKNESS_M
MASS = RHO_AL6061 * V
HEAT_CAPACITY = MASS * CP_AL6061
ENERGY_J = HEAT_CAPACITY * (T_SET_C - T_AMBIENT_C)
AVERAGE_RAMP_POWER_W = ENERGY_J / (RAMP_MIN * 60)

T_HOT_K = T_SET_C + 273.15
T_AMB_K = T_AMBIENT_C + 273.15
Q_RAD_PER_M2 = EMISSIVITY * SIGMA * (T_HOT_K**4 - T_AMB_K**4)
Q_CONV_PER_M2 = H_CONVECTION * (T_SET_C - T_AMBIENT_C)

TOP_LOSS_W = A_TOP * (Q_RAD_PER_M2 + Q_CONV_PER_M2)
SIDE_LOSS_W = A_SIDE * (Q_RAD_PER_M2 + Q_CONV_PER_M2)
BOTTOM_LOSS_W = (INSULATION_K / INSULATION_THICKNESS_M) * A_TOP * (T_SET_C - T_AMBIENT_C)
STANDOFF_LOSS_W = 5.0  # placeholder estimate; depends on geometry/material
TOTAL_LOSS_W = TOP_LOSS_W + SIDE_LOSS_W + BOTTOM_LOSS_W + STANDOFF_LOSS_W

# Approximate radial temperature drop for a disk with side loss.
# This is only a simplified conduction estimate and not a uniformity prediction.
R_WAFER = WAFER_DIAMETER_M / 2
RADIAL_DROP_C = SIDE_LOSS_W / (2 * math.pi * K_AL6061 * THICKNESS_M) * math.log(R / R_WAFER)

wafer_volume = math.pi * R_WAFER**2 * WAFER_THICKNESS_M
wafer_mass = RHO_SILICON * wafer_volume
wafer_heat_capacity = wafer_mass * CP_SILICON
```

```
print("FIRST-ORDER HOT-PLATE THERMAL SIZING — NOT VALIDATED PERFORMANCE")
print(f"Plate: {DIAMETER_M*1000:.0f} mm diameter x {THICKNESS_M*1000:.0f} mm")
print(f"Volume: {V*1e6:.1f} cm^3")
print(f"Plate mass: {MASS:.3f} kg")
print(f"Plate heat capacity: {HEAT_CAPACITY:.1f} J/K")
print(f"Energy from {T_AMBIENT_C:.0f} C to {T_SET_C:.0f} C: {ENERGY_J/1000:.1f} kJ")
print(f"Average ideal power for {RAMP_MIN:.0f}-minute ramp (ignores losses): {AVERAGE_RAMP_POWER_W:.1f} W")
print(f"Estimated losses at {T_SET_C:.0f} C: top={TOP_LOSS_W:.1f} W, side={SIDE_LOSS_W:.1f} W,
bottom={BOTTOM_LOSS_W:.1f} W, standoffs={STANDOFF_LOSS_W:.1f} W")
print(f"Total estimated steady-state loss: {TOTAL_LOSS_W:.1f} W")
print(f"Approximate radial drop from wafer edge (r={R_WAFER*1000:.1f} mm) to plate edge: {RADIAL_DROP_C:.2f} C
(simplified model only)")
print(f"4-inch silicon wafer estimate: mass={wafer_mass*1000:.1f} g, heat capacity={wafer_heat_capacity:.1f} J/K")
print("Installed heater power: 1,000 W nominal (4 x 250 W); controller must limit ramp and prevent overshoot.")
print("Do not claim ±0.5 C uniformity or repeatability until the validation plan passes.")
```

The script performs four main tasks:

1. Calculate plate dimensions and mass: Python calculates the volume of a cylindrical plate: $V = \pi r^2 h$
   Then it uses density to calculate mass: $m = \rho V$
2. Calculate the energy required: The energy needed to heat the plate is estimated using: $Q = mc_p\Delta T$
   This gives approximately 364.8 kJ for heating from 25°C to 400°C.
3. Estimate heater power: For a 20-minute ideal heating ramp: $P_{\text{avg}} = \frac{Q}{t}$
   This gives approximately 304 W before heat losses.
4. Estimate heat losses: The code estimates heat lost through convection, radiation, insulation and supports. These are approximate model results, not measurements from a real hot plate.

```
PS C:\Users\Lenovo\Downloads\SureshKumarK_HackerFabIITB_Submission> py calcs/ther
mal_calc.py
FIRST-ORDER HOT-PLATE THERMAL SIZING — NOT VALIDATED PERFORMANCE
Plate: 160 mm diameter x 20 mm
Volume: 402.1 cm^3
Plate mass: 1.086 kg
Plate heat capacity: 972.8 J/K
Energy from 25 C to 400 C: 364.8 kJ
Average ideal power for 20-minute ramp (ignores losses): 304.0 W
Estimated losses at 400 C: top=105.3 W, side=52.7 W, bottom=9.0 W, standoffs=5.0
W
Total estimated steady-state loss: 172.1 W
Approximate radial drop from wafer edge (r=50.8 mm) to plate edge: 1.14 C (simpli
fied model only)
4-inch silicon wafer estimate: mass=9.9 g, heat capacity=6.9 J/K
Installed heater power: 1,000 W nominal (4 x 250 W); controller must limit ramp a
nd prevent overshoot.
Do not claim ±0.5 C uniformity or repeatability until the validation plan passes.
```

thermal_results.txt

```
FIRST-ORDER HOT-PLATE THERMAL SIZING ù NOT VALIDATED PERFORMANCE
Plate: 160 mm diameter x 20 mm
Volume: 402.1 cm^3
Plate mass: 1.086 kg
Plate heat capacity: 972.8 J/K
Energy from 25 C to 400 C: 364.8 kJ
Average ideal power for 20-minute ramp (ignores losses): 304.0 W
Estimated losses at 400 C: top=105.3 W, side=52.7 W, bottom=9.0 W, standoffs=5.0 W
Total estimated steady-state loss: 172.1 W
Approximate radial drop from wafer edge (r=50.8 mm) to plate edge: 1.14 C (simplified model only)
4-inch silicon wafer estimate: mass=9.9 g, heat capacity=6.9 J/K
Installed heater power: 1,000 W nominal (4 x 250 W); controller must limit ramp and prevent overshoot.
Do not claim ±0.5 C uniformity or repeatability until the validation plan passes.
```

Quantity	Calculated result
Energy to heat plate from 25°C to 400°C	364.8 kJ
Ideal average power for a 20-minute ramp	304 W
Estimated heat loss at 400°C	172.1 W
Approximate ramp-plus-loss demand	476 W
Installed heater power	1,000 W

The calculations suggest that 1 kW of installed heating power is a reasonable starting point. They do not prove the actual heating time or temperature performance, because real heater contact, thermal losses, and controller behaviour still need testing.

The hardest requirement is not reaching 400°C. It is achieving all three of these simultaneously:

- Temperature uniformity across the wafer within ±0.5°C.
- Run-to-run repeatability below 0.5°C.
- Surface flatness within ±10 µm.

The proposed solution uses separate centre and edge heater zones, calibrated sensors, a fixed heating-and-soaking protocol, and measurements across the wafer footprint.

However, aluminium 6061-T6 can lose its original temper and distort during high-temperature exposure. The proposed design therefore requires thermal cycling and flatness measurements after machining. These performance specifications remain unverified until a prototype passes the tests.

CAD design: Using FreeCAD

HackerFab_Hot_Plate_AllInOne.FCMacro

```
# HackerFab hot-plate concept assembly for FreeCAD 1.1+
# Units: mm. This is a preliminary explanatory CAD model, NOT a manufacturing release.
import FreeCAD as App
import Part
import math
from FreeCAD import Vector

try:
if App.ActiveDocument:
App.closeDocument(App.ActiveDocument.Name)
except Exception:
pass

doc = App.newDocument("HackerFab_Hot_Plate_Concept")

# Colors are only for visual differentiation.
ALUMINIUM = (0.78, 0.80, 0.83)
HEATER = (0.88, 0.34, 0.08)
SENSOR = (0.12, 0.45, 0.78)
INSULATION = (0.83, 0.78, 0.62)
STEEL = (0.45, 0.48, 0.51)
DARK = (0.20, 0.22, 0.24)
WAFER = (0.30, 0.72, 0.88)


def feature(name, label, shape, color, transparency=0):
obj = doc.addObject("PartDesign::Feature", name)
obj.Label = label
obj.Shape = shape
if hasattr(obj, "ViewObject"):
obj.ViewObject.ShapeColor = color
obj.ViewObject.LineColor = (0.12, 0.12, 0.12)
obj.ViewObject.DisplayMode = "Flat Lines"
obj.ViewObject.Transparency = transparency
return obj

# Main plate: top surface at z=20, bottom surface at z=0.
plate_blank = Part.makeCylinder(80.0, 20.0, Vector(0, 0, 0), Vector(0, 0, 1))

# Four radial cartridge-heater bores, entered from the rim. This is physically
# consistent with a 20 mm plate thickness and 60 mm long heater cartridges.
# Hole axis lies at mid-thickness z=10; bores run radially inward 60 mm.
heater_bore_shapes = []
heater_directions = [Vector(-1,0,0), Vector(1,0,0), Vector(0,-1,0), Vector(0,1,0)]
heater_starts = [Vector(80,0,10), Vector(-80,0,10), Vector(0,80,10), Vector(0,-80,10)]
for start, direction in zip(heater_starts, heater_directions):
heater_bore_shapes.append(Part.makeCylinder(3.25, 60.0, start, direction))
for bore in heater_bore_shapes:
plate_blank = plate_blank.cut(bore)

# Two underside RTD pockets, diameter 3.2 mm, 14 mm deep. Kept away from radial bores.
for x, y in [(-28,-28), (28,28)]:
pocket = Part.makeCylinder(1.6, 14.0, Vector(x,y,0), Vector(0,0,1))
plate_blank = plate_blank.cut(pocket)
plate_obj = feature("HeatingPlate", "01 - Aluminium heating plate (Ø160 × 20)", plate_blank, ALUMINIUM)
plate_obj.addProperty("App::PropertyLength", "Diameter", "Design").Diameter = 160
plate_obj.addProperty("App::PropertyLength", "Thickness", "Design").Thickness = 20
plate_obj.addProperty("App::PropertyString", "Material", "Design").Material = "Aluminium 6061 - provisional; verify high-
temperature strength/flatness"

# Four cartridge heaters, modeled as solid cylinders inserted radially. Their
# nominal length is 60 mm and diameter 6.0 mm for a little visual clearance.
for i, (start, direction) in enumerate(zip(heater_starts, heater_directions), start=1):
# Begin 1 mm outside the rim and run inward 60 mm.
heater_start = start + direction * (-1.0)
heater_shape = Part.makeCylinder(3.0, 60.0, heater_start, direction)
h = feature("CartridgeHeater%02d" % i, "%02d - Cartridge heater %d (concept Ø6 × 60)" % (i+1, i), heater_shape, HEATER)
h.addProperty("App::PropertyString", "Rating", "Design").Rating = "250 W nominal; confirm supplier and voltage"

# RTD probes are represented as vertical cylinders entering from the underside.
for i, (x, y) in enumerate([(-28,-28), (28,28)], start=1):
sensor_shape = Part.makeCylinder(1.35, 14.0, Vector(x,y,0), Vector(0,0,1))
```

```
s = feature("Pt100Sensor%02d" % i, "%02d - Pt100 sensor %d (concept Ø2.7 × 14)" % (i+5, i), sensor_shape, SENSOR)
s.addProperty("App::PropertyString", "Type", "Design").Type = "4-wire Pt100; pocket fit and response require datasheet
check"

# Wafer reference: thin translucent disk sitting on top. It is a visual reference,
# not a bonded component.
wafer_shape = Part.makeCylinder(50.8, 0.5, Vector(0,0,20.2), Vector(0,0,1))
waf = feature("WaferReference", "08 - 4-inch wafer reference (Ø101.6; transparent)", wafer_shape, WAFER, 72)
waf.addProperty("App::PropertyLength", "Diameter", "Design").Diameter = 101.6

# Insulation disk directly below the plate (schematic dimensions).
ins_shape = Part.makeCylinder(90.0, 25.0, Vector(0,0,-25.0), Vector(0,0,1))
ins = feature("Insulation", "09 - Ceramic-fibre insulation disk (Ø180 × 25; concept)", ins_shape, INSULATION)

# Four support spacers under insulation.
for i, (x,y) in enumerate([(-65,-65),(-65,65),(65,-65),(65,65)], start=1):
spacer_shape = Part.makeCylinder(5.0, 18.0, Vector(x,y,-43.0), Vector(0,0,1))
feature("SupportSpacer%02d" % i, "Support spacer %d (concept Ø10 × 18)" % i, spacer_shape, STEEL)

# Simple square enclosure/base. It is schematic and intentionally separate from the hot plate.
base_outer = Part.makeBox(190.0, 190.0, 18.0, Vector(-95,-95,-65))
base_inner = Part.makeBox(178.0, 178.0, 18.2, Vector(-89,-89,-65.1))
base_shape = base_outer.cut(base_inner)
feature("BaseEnclosure", "14 - Stainless enclosure base (schematic 190 × 190 × 18)", base_shape, STEEL)

# Four feet below base.
for i, (x,y) in enumerate([(-72,-72),(-72,72),(72,-72),(72,72)], start=1):
foot_shape = Part.makeCylinder(8.0, 10.0, Vector(x,y,-75), Vector(0,0,1))
feature("Foot%02d" % i, "Enclosure foot %d (schematic Ø16 × 10)" % i, foot_shape, DARK)

# A front-panel representation on the front side (negative Y): controller, indicator, E-stop.
panel = Part.makeBox(70, 3, 32, Vector(-35,-98,-62))
feature("FrontPanel", "19 - Front control panel (schematic)", panel, DARK)
controller = Part.makeBox(28, 2, 18, Vector(-25,-100,-56))
feature("ControllerDisplay", "20 - PID display (schematic)", controller, (0.12,0.16,0.18))
for i, x in enumerate([15, 28], start=1):
knob = Part.makeCylinder(4.0, 3.0, Vector(x,-101,-48), Vector(0,1,0))
feature("PanelControl%02d" % i, "Panel control %d (schematic)" % i, knob, (0.10,0.65,0.30) if i == 1 else (0.78,0.08,0.08))

# Organize objects into groups for a clean model tree.
# Feature objects remain editable as individual Part::Feature solids.
try:
assembly = doc.addObject("App::DocumentObjectGroup", "Assembly")
assembly.Label = "Hot plate assembly - concept components"
for obj in list(doc.Objects):
if obj != assembly and obj.Name != "Origin":
assembly.addObject(obj)
except Exception:
pass

doc.recompute()
Gui.activeDocument().activeView().viewAxonometric()
Gui.activeDocument().activeView().fitAll()

# Set a neutral background/display style if available.
try:
Gui.activeDocument().activeView().setAnimationEnabled(False)
except Exception:
pass

App.Console.PrintMessage("\nHot plate concept model created. Save as Hot_Plate_Assembly.FCStd.\n")
App.Console.PrintMessage("IMPORTANT: concept geometry only; heater bore arrangement, sensor pockets, enclosure and
material must be validated before manufacture.\n")


# ------------------------------------------------------------------------------
# AUTOMATIC DIMENSIONED DRAWING SHEET (FreeCAD 1.1.x)
# ------------------------------------------------------------------------------
# A4 landscape TechDraw page, plate top/front views, nominal diameter dimension,
# and a requirements / safety / BOM / assembly note block. This is a CONCEPT
# design; requirements are targets and are NOT claimed as achieved/tested.

import os

try:
# Find a bundled A4 landscape TechDraw template from this FreeCAD install.
```

```
template_candidates = [
App.getResourceDir() + "Mod/TechDraw/Templates/A4_LandscapeTD.svg",
App.getResourceDir() + "Mod/TechDraw/Templates/A4_Landscape.svg",
App.getResourceDir() + "Mod/TechDraw/Templates/A4_Landscape_blank.svg",
]
template_path = next((p for p in template_candidates if os.path.exists(p)), None)

page = doc.addObject("TechDraw::DrawPage", "DimensionedDrawingPage")
page.Label = "Hot Plate - Dimensioned Concept Drawing"
if template_path:
template = doc.addObject("TechDraw::DrawSVGTemplate", "DimensionedDrawingTemplate")
template.Label = "A4 Landscape Drawing Sheet"
template.Template = template_path
page.Template = template

# Top view: plate only, so heater bores and sensor pockets can be inspected.
top = doc.addObject("TechDraw::DrawViewPart", "PlateTopView")
top.Label = "Heating plate - top view"
top.Source = [plate_obj]
top.Direction = Vector(0, 0, 1)
top.ScaleType = "Custom"
top.Scale = 0.62
top.X = 76
top.Y = 155
page.addView(top)

# Front view: plate only, communicates the 20 mm nominal thickness.
front = doc.addObject("TechDraw::DrawViewPart", "PlateFrontView")
front.Label = "Heating plate - front view"
front.Source = [plate_obj]
front.Direction = Vector(0, -1, 0)
front.ScaleType = "Custom"
front.Scale = 0.62
front.X = 76
front.Y = 88
page.addView(front)

# Isometric view of the concept assembly for context.
assembly_view = doc.addObject("TechDraw::DrawViewPart", "AssemblyIsometricView")
assembly_view.Label = "Concept assembly - isometric"
assembly_view.Source = [o for o in doc.Objects if hasattr(o, "Shape") and not o.Shape.isNull() and o.Name not in
("HeatingPlate",)]
assembly_view.Direction = Vector(1, -1, 1)
assembly_view.ScaleType = "Custom"
assembly_view.Scale = 0.28
assembly_view.X = 190
assembly_view.Y = 137
page.addView(assembly_view)

doc.recompute()

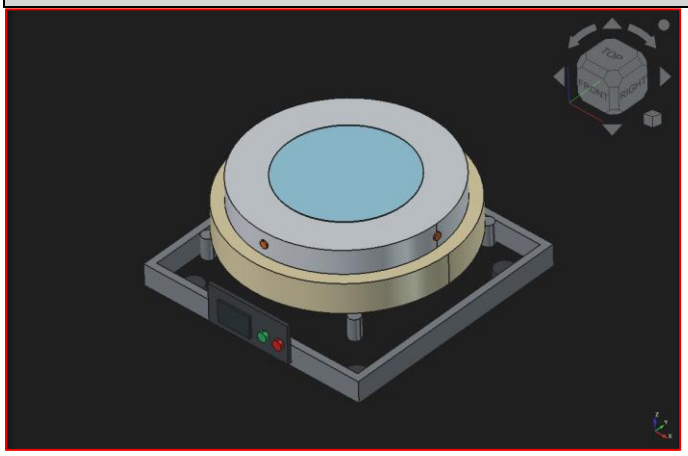
# Find the top circular edge of the plate for a real TechDraw diameter dim.
outer_edge_name = None
outer_edge_z = -1e99
for i, edge in enumerate(plate_obj.Shape.Edges, start=1):
try:
curve = edge.Curve
if hasattr(curve, "Radius") and abs(float(curve.Radius) - 80.0) < 0.05:
z = edge.CenterOfMass.z
if z > outer_edge_z:
outer_edge_z = z
outer_edge_name = "Edge%d" % i
except Exception:
pass

if outer_edge_name:
try:
diameter_dim = doc.addObject("TechDraw::DrawViewDimension", "PlateDiameterDimension")
diameter_dim.Label = "Plate outside diameter - Ø160 mm"
diameter_dim.Type = "Diameter"
diameter_dim.MeasureType = "True"
diameter_dim.References3D = [(plate_obj, [outer_edge_name])]
diameter_dim.X = 105
diameter_dim.Y = 125
page.addView(diameter_dim)
except Exception as e:
```

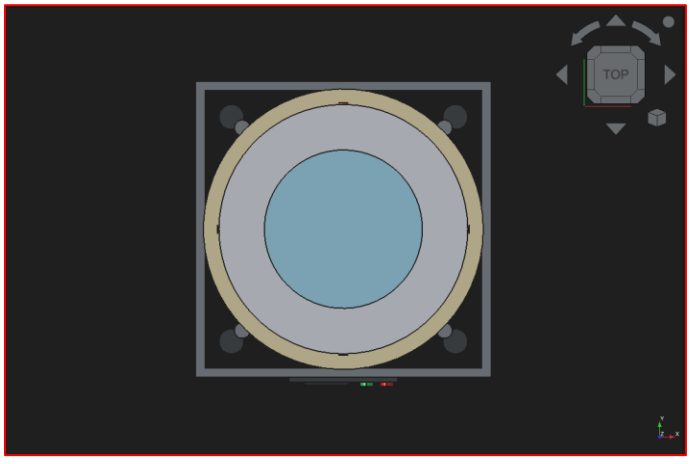
```
App.Console.PrintWarning("Automatic diameter dimension could not be attached: %s\n" % str(e))

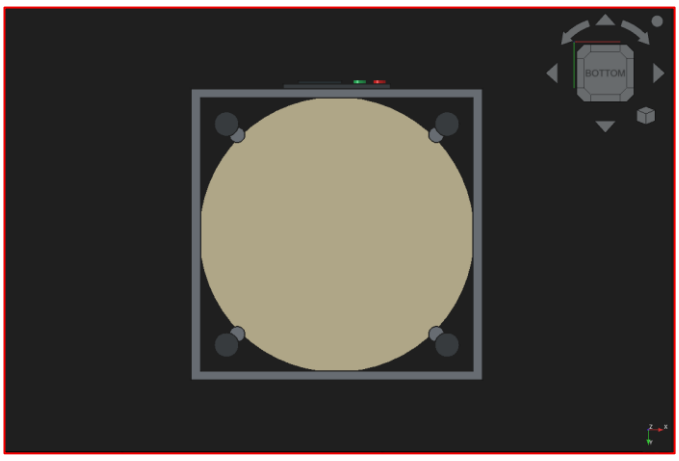
# Notes block includes dimensional callouts and all key task requirements.
# Text uses explicit target language so no unverified performance is implied.
notes = doc.addObject("TechDraw::DrawViewAnnotation", "DesignRequirementsNotes")
notes.Label = "Dimensions, performance targets, BOM and assembly notes"
notes.Text = [
"HACKERFAB IITB - SEMICONDUCTOR HOT PLATE CONCEPT",
"UNITS: mm unless stated | PRELIMINARY - NOT FOR MANUFACTURE",
"GEOMETRY: plate Ø160 x 20 thick; active wafer reference Ø101.6 (4 inch)",
"HEATING: 4 x 250 W nominal cartridge heaters, concept Ø6 x 60 radial",
"SENSING: 2 x 4-wire Pt100 RTDs; positions/pockets to be validated",
"TEMPERATURE RANGE TARGET: ambient to 400 deg C",
"SETPOINT ERROR TARGET: <5 deg C; run-to-run difference target: <0.5 deg C",
"UNIFORMITY TARGET: +/-0.5 deg C over qualified active area (max spread 1 deg C)",
"FLATNESS TARGET: +/-10 micrometres across top surface",
"CONTROL/SAFETY: PID + SSR/contactor; independent over-temperature cutoff and fuse",
"BOM (concept): Al plate; 4 heaters; 2 Pt100 probes; PID/controller; SSR; contactor; fuse;",
"independent thermal cutout; insulation; enclosure; wiring, terminals and earth hardware",
"ASSEMBLY: machine/inspect plate and bores; fit heaters and RTDs; add insulation/base;",
"wire earth, fuse, contactor, SSR and independent cutoff; verify sensor polarity and interlocks",
"QUALIFICATION REQUIRED: map temperature at multiple points, repeat cycles, verify flatness",
"and stability at 400 deg C. Do not claim targets met until measured and documented.",
"MATERIAL CAUTION: 6061 aluminium temper/strength and distortion at 400 deg C require review.",
"All heater bores, clearances, electrical ratings, insulation and guarding require engineering sign-off."
]
notes.X = 190
notes.Y = 55
notes.TextSize = 2.15
page.addView(notes)

doc.recompute()
Gui.activeDocument().activeView().viewAxonometric()
Gui.activeDocument().activeView().fitAll()
App.Console.PrintMessage("\nCreated concept assembly AND dimensioned TechDraw sheet.\n")
App.Console.PrintMessage("Inspect the TechDraw page and confirm all views/annotations are visible before submission.\n")
App.Console.PrintMessage("Performance values are design targets, not experimentally verified results.\n")
except Exception as drawing_error:
App.Console.PrintError("Assembly was created, but drawing-sheet creation encountered an error: %s\n" % str(drawing_error))
App.Console.PrintMessage("The 3D concept assembly remains in the document. Check the Report view for the error.\n")
```









Hot_Plate_Drawing_FINAL.FCMacro

```
# Self-contained concept drawing macro for FreeCAD 1.1.x.
# Creates a fresh TechDraw page with a vector SVG sheet. Keeps all 3D objects intact.
import FreeCAD as App
import os, traceback

if App.ActiveDocument is None:
raise RuntimeError('Open your hot-plate 3D document first, then run this macro.')
doc = App.ActiveDocument

svg = '''<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<svg xmlns="http://www.w3.org/2000/svg" width="297mm" height="210mm" viewBox="0 0 297 210">
<rect x="5" y="5" width="287" height="200" fill="white" stroke="black" stroke-width="0.35"/><rect x="9" y="9" width="279"
height="192" fill="none" stroke="black" stroke-width="0.18"/>
<g fill="black" font-family="DejaVu Sans,Arial,sans-serif">
<text x="15" y="17" font-size="3.6" font-weight="bold">HACKERFAB IITB | SEMICONDUCTOR HOT PLATE</text><text x="15" y="22"
font-size="2.5">PRELIMINARY CONCEPT DRAWING - NOT FOR MANUFACTURE</text>
<text x="45" y="32" font-size="3" font-weight="bold" text-anchor="middle">TOP VIEW</text><circle cx="45" cy="66" r="25"
fill="none" stroke="black" stroke-width="0.45"/><circle cx="45" cy="66" r="15.875" fill="none" stroke="black" stroke-
width="0.35" stroke-dasharray="1.5,1"/>
<circle cx="45" cy="43" r="1.2" fill="none" stroke="black" stroke-width="0.35"/><circle cx="68" cy="66" r="1.2"
fill="none" stroke="black" stroke-width="0.35"/><circle cx="45" cy="89" r="1.2" fill="none" stroke="black" stroke-
width="0.35"/><circle cx="22" cy="66" r="1.2" fill="none" stroke="black" stroke-width="0.35"/>
<line x1="20" y1="98" x2="70" y2="98" stroke="black" stroke-width="0.25"/><line x1="20" y1="94" x2="20" y2="100"
stroke="black" stroke-width="0.25"/><line x1="70" y1="94" x2="70" y2="100" stroke="black" stroke-width="0.25"/><path
d="M20 98 l3 -1.2 v2.4 z M70 98 l-3 -1.2 v2.4 z" fill="black"/><text x="45" y="96" font-size="2.7" text-anchor="middle">Ø160
mm</text>
<text x="45" y="105" font-size="2.3" text-anchor="middle">Dashed circle: wafer reference Ø101.6</text><text x="45" y="110"
font-size="2.3" text-anchor="middle">Four heater positions are schematic</text>
<text x="45" y="123" font-size="3" font-weight="bold" text-anchor="middle">FRONT / EDGE VIEW</text><rect x="20" y="133"
width="50" height="6.25" fill="none" stroke="black" stroke-width="0.45"/><line x1="20" y1="143" x2="70" y2="143"
stroke="black" stroke-width="0.25"/><line x1="20" y1="141" x2="20" y2="145" stroke="black" stroke-width="0.25"/><line
x1="70" y1="141" x2="70" y2="145" stroke="black" stroke-width="0.25"/><text x="45" y="148" font-size="2.7" text-
anchor="middle">160 mm</text><line x1="75" y1="133" x2="75" y2="139.25" stroke="black" stroke-width="0.25"/><line x1="73"
y1="133" x2="77" y2="133" stroke="black" stroke-width="0.25"/><line x1="73" y1="139.25" x2="77" y2="139.25" stroke="black"
stroke-width="0.25"/><text x="79" y="137" font-size="2.5">20 mm</text><text x="45" y="155" font-size="2.2" text-
anchor="middle">Schematic side profile; verify detailed design</text>
<text x="105" y="32" font-size="3.1" font-weight="bold">DESIGN BASIS AND REQUIREMENTS</text><text x="105" y="39" font-
size="2.55">Plate: nominal Ø160 x 20 mm; wafer reference Ø101.6 mm (4 inch).</text><text x="105" y="45" font-
size="2.55">Heating: 4 x nominal 250 W cartridge heaters, concept Ø6 x 60 mm.</text><text x="105" y="51" font-
size="2.55">Sensing: 2 x four-wire Pt100 RTDs; locations need validation.</text><text x="105" y="57" font-
size="2.55">Operating target: ambient to 400 °C.</text><text x="105" y="63" font-size="2.55">Setpoint error target: less
than 5 °C.</text><text x="105" y="69" font-size="2.55">Run-to-run difference target: less than 0.5 °C.</text><text x="105"
y="75" font-size="2.55">Uniformity target: ±0.5 °C in qualified area; max spread 1 °C.</text><text x="105" y="81" font-
size="2.55">Top flatness target: ±10 µm; verify by suitable metrology.</text>
<text x="105" y="91" font-size="3.1" font-weight="bold">BILL OF MATERIALS - CONCEPTUAL</text><text x="105" y="98" font-
size="2.55">1 x aluminium plate; alloy/temper must be qualified at 400 °C.</text><text x="105" y="104" font-size="2.55">4
x cartridge heaters; 2 x four-wire Pt100 sensors.</text><text x="105" y="110" font-size="2.55">PID controller; rated SSR;
contactor; fuse or circuit breaker.</text><text x="105" y="116" font-size="2.55">Independent over-temperature cutoff;
rated wires and terminals.</text><text x="105" y="122" font-size="2.55">Thermal insulation; grounded enclosure; supports
and fasteners.</text>
<text x="105" y="132" font-size="3.1" font-weight="bold">ASSEMBLY AND SAFETY NOTES</text><text x="105" y="139" font-
size="2.55">1. Review alloy, heater fit, bore layout and high-temperature strength.</text><text x="105" y="145" font-
size="2.55">2. Fit heaters and RTDs; provide strain relief and thermal insulation.</text><text x="105" y="151" font-
size="2.55">3. Bond protective earth; use rated switching and independent cutoff.</text><text x="105" y="157" font-
size="2.55">4. Guard live terminals; verify wiring, insulation and grounding.</text><text x="105" y="163" font-
size="2.55">5. Map temperatures at multiple points; repeat cycles; measure flatness.</text>
<text x="105" y="173" font-size="2.8" font-weight="bold">QUALIFICATION LIMITATIONS</text><text x="105" y="180" font-
size="2.5">Targets are not demonstrated until tested. Aluminium 6061 at 400 °C</text><text x="105" y="186" font-
size="2.5">requires review for temper loss, warpage and flatness retention.</text><text x="105" y="192" font-
size="2.5">Concept-level aid only; not a released manufacturing drawing.</text>
<line x1="9" y1="195" x2="288" y2="195" stroke="black" stroke-width="0.25"/><text x="15" y="200" font-size="2.3">UNITS:
mm | SCALE: schematic | REV: A | DATE: 2026-10-13</text><text x="280" y="200" font-size="2.3" text-anchor="end">HACKERFAB
IITB TASK</text></g></svg>'''

svg_path = os.path.join(os.path.expanduser('~'), 'Hot_Plate_Dimensioned_Concept.svg')
with open(svg_path, 'w', encoding='utf-8') as f:
f.write(svg)

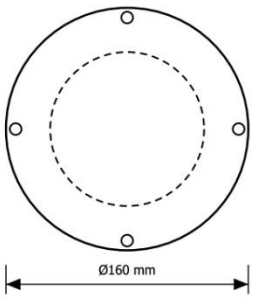
try:
# New uniquely named page/template; existing 3D geometry and old page are untouched.
template = doc.addObject('TechDraw::DrawSVGTemplate', 'FinalHotPlateTemplate')
template.Label = 'A4 Landscape - Dimensioned Hot Plate Concept'
template.Template = svg_path
```

```
page = doc.addObject('TechDraw::DrawPage', 'FinalHotPlateDrawingPage')
page.Label = 'FINAL - Hot Plate Dimensioned Concept Drawing'
page.Template = template
doc.recompute()
Gui.activateWorkbench('TechDrawWorkbench')
Gui.activeDocument().setEdit(page.Name)
doc.recompute()
App.Console.PrintMessage('\nSUCCESS: Created a new self-contained concept drawing page. If not visible, double-click FINAL
- Hot Plate Dimensioned Concept Drawing in the tree.\nSVG file: ' + svg_path + '\n')
App.Console.PrintMessage('3D components were not deleted or modified. Save the document.\n')
except Exception as e:
App.Console.PrintError('DRAWING CREATION FAILED: %s\n%s\n' % (e, traceback.format_exc()))
raise
```

HACKERFAB IITB | SEMICONDUCTOR HOT PLATE

PRELIMINARY CONCEPT DRAWING - NOT FOR MANUFACTURE

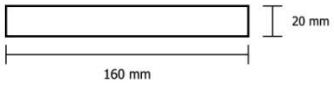
TOP VIEW



Dashed circle: wafer reference Ø101.6

Four heater positions are schematic

FRONT / EDGE VIEW



Schematic side profile; verify detailed design

DESIGN BASIS AND REQUIREMENTS

Plate: nominal Ø160 x 20 mm; wafer reference Ø101.6 mm (4 inch).

Heating: 4 x nominal 250 W cartridge heaters, concept Ø6 x 60 mm.

Sensing: 2 x four-wire Pt100 RTDs; locations need validation.

Operating target: ambient to 400 °C.

Setpoint error target: less than 5 °C.

Run-to-run difference target: less than 0.5 °C.

Uniformity target: ±0.5 °C in qualified area; max spread 1 °C.

Top flatness target: ±10 µm; verify by suitable metrology.

BILL OF MATERIALS - CONCEPTUAL

1 x aluminium plate; alloy/temper must be qualified at 400 °C.

4 x cartridge heaters; 2 x four-wire Pt100 sensors.

PID controller; rated SSR; contactor; fuse or circuit breaker.

Independent over-temperature cutoff; rated wires and terminals.

Thermal insulation; grounded enclosure; supports and fasteners.

ASSEMBLY AND SAFETY NOTES

1. Review alloy, heater fit, bore layout and high-temperature strength.
2. Fit heaters and RTDs; provide strain relief and thermal insulation.
3. Bond protective earth; use rated switching and independent cutoff.
4. Guard live terminals; verify wiring, insulation and grounding.
5. Map temperatures at multiple points; repeat cycles; measure flatness.

QUALIFICATION LIMITATIONS

Targets are not demonstrated until tested. Aluminium 6061 at 400 °C

requires review for temper loss, warpage and flatness retention.

Concept-level aid only; not a released manufacturing drawing.

UNITS: mm | SCALE: schematic | REV: A | DATE: 2026-10-13

HACKERFAB IITB TASK